

# Joseph Matthew Rodriguez Ruelas

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## SKILLS

Python (PyTorch, Tensorflow, Scikit-learn, Pandas, etc.) • Automation • Machine Learning • AWS • SQL • Kubernetes • Docker • Uncertainty Quantification • Data Analysis • HPC • Data Engineering • Jira • Agile Methodology • Git

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## SUMMARY

Astrophysics student at UC San Diego specializing in machine learning, uncertainty quantification, and large-scale scientific inference. Experienced in deploying ML pipelines across research and infrastructure environments, with prior industry experience at General Atomics automating cloud/on-prem systems. Seeking ML or data-driven roles in scientific and industrial applications.

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## WORK EXPERIENCE

*School of Physical Sciences | UC San Diego, **Undergraduate Researcher***

*Apr 2025 - Present*

- Developed a physics-informed deep learning setup in Python, combining multiple population-synthesis codes (COMPAS, COSMIC, POSYDON) to run joint simulation-based inference, enhancing model accuracy.
- Designed and implemented an end-to-end inference pipeline using CNNs/Transformers and normalizing flows, improving efficiency in generating and analyzing large synthetic populations by 80%.
- Created tools for cross-code comparisons and interpretability checks, identifying model agreement and breakdowns, leading to more robust astrophysical formation models.

*General Atomics, **Infrastructure Systems Intern***

*Jun 2025 - Aug 2025*

- Built and managed end-to-end server environments across cloud and on-prem (Windows/Linux VMs, AWS instances), handling full lifecycle from provisioning to decommissioning, plus hands-on data center work.
- Developed automation scripts in PowerShell, Microsoft Orchestrator, and Python, cutting troubleshooting and audit time by up to 95% by identifying orphaned servers and generating compliance reports.
- Integrated AI tools into Infrastructure Operations, leading training sessions and demos to show interns and employees how AI could be used in real operational workflows.

*SPACES at UC San Diego, **Academic Success Program Coordinator***

*Jun 2024 - Jun 2025*

- Managed UCSD's largest equity-focused textbook program, supporting ~900 students annually and overseeing inventory, logistics, and a \$55K budget, ensuring efficient operations during peak periods.
  - Developed and enhanced backend analytics using Python and spreadsheets, improving application processing speed by 80% and enabling smarter purchasing decisions.
  - Led a 100+ student survey on course-material costs, presented findings to the UCSD Academic Senate, and contributed to expanding Inclusive Access support.
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## PROJECT EXPERIENCE

[Coastal Flood Prediction System](#): GRU-based time-series ML, NOAA tide-gauge data

[ROI Analysis of Renewable Energy Transition Across U.S. States](#): Python, long-horizon economic modeling

[Gas Morphology Classifier for Binary Evolution](#): CNN-based astrophysical simulation analysis

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## EDUCATION

*UC San Diego, Bachelor of Science • **Astrophysical Sciences***

*Expected June 2027*

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## HONORS & AWARDS

*Hispanic Scholarship Fund • **HSF Scholar***